Anubhav Agrawal

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Technical Skills

Languages & Tools: Python, Bash, Git, Linux
Model Optimization: PyTorch, ONNX (dynamic shapes, control flow, opsets), ONNX Runtime, TensorRT, trtexec
Quantization & On-Device: INT8/FP16 PTQ with calibration, 4-bit GGUF/MLX (llama.cpp, mlx-lm)
ML Frameworks: PyTorch, OpenCV, Scikit-Learn, XGBoost, NumPy, Pandas
Platform: Docker, GitLab CI/CD, AWS (EC2), FastAPI, Grafana/Loki/Mimir/Tempo, MongoDB, SQLite

Experience

Griphic *Delhi*
AI Engineer II *Jan 2026 – Present*

- Sole engineer on a multi-agent LLM service in production (Python/FastAPI) that turns interviews into structured blueprints, end to end.
- Cut per-request generation cost by **60%** by tracking what every token costs, caching prompts, and bounding parallel fan-out.
- Made outputs safe to ship unread: schema-validated structured outputs with bounded self-correction (**99.5%** ship with zero human review), plus a two-stage guardrail — batched judge, then independent adjudicator — that stops prompt injection and credential leaks for at most **2 extra LLM calls** per clean request.

Innvolution Healthcare Pvt. Ltd. *Bangalore*
Machine Learning Engineer *& Team Lead* *Sep 2023 – Dec 2025*

- Owned AIMAG, an X-ray super-resolution and denoising product at **$200K+ ARR**, from scratch to serving — datasets, model R&D, quantization, and release after clinical trials.
- Made the model real-time: PyTorch → ONNX → TensorRT with INT8/FP16 **post-training quantization**, debugging dynamic shapes, unsupported ops, and opset mismatches in export; **sub-5 ms per frame** for 4× upsampling on an RTX 4090, quantized accuracy checked against the FP32 baseline.
- Built the QCA pipeline (segmentation → calibration → prediction) on a fine-tuned U-Net++ (ResNet50 + attention); live cardiac-catheterization analysis completes in under 4 seconds.
- Co-inventor on a **granted Indian patent** (No. 604176, AI-Powered X-Ray Image Enhancement) and **2 pending applications**: blind single-image super-resolution, DSA roadmap generation, and real-time fluoroscopy enhancement.
- Led **6 ML engineers**: owned the technical roadmap, set up architecture and code review, and wrote each release's deployment runbook (calibration steps, accuracy gates, latency budgets).

Projects

ipic — On-Device Multimodal Search Engine *GitHub | Write-ups*
Fully offline search across text, PDFs, images, audio, and video — no cloud, no telemetry *2026*

- Architected a sharded index — one self-contained SQLite + vector-store shard per root — under a supervising engine that parallelizes scan, embedding, and search; BGE and CLIP run entirely on-device.
- Fused dense (bge-small), CLIP vision, BM25 (FTS5), and acoustic lanes via reciprocal rank fusion over custom mmap'd i8-quantized vector stores; exact scan beats HNSW at file-index scale: **26–32 ms** queries, sub-millisecond vector scans.
- Scaled to a **350k-file** home directory at **∼2.5k files/s** with bounded-queue backpressure, batched shard claims, and a per-stage compute budget — eliminating the SQLite convoy and queue growth that capped v1.

Inference Lab — LLM Serving Benchmarks on Apple Silicon *GitHub | Write-up*
Reproducible benchmark harness: mlx-lm vs llama.cpp under a single fairness contract *2026*

- Built a reproducible benchmark harness holding mlx-lm and llama.cpp to one fairness contract (fixed prompts, greedy decoding, matched budgets): **45–56 tok/s** single-stream for Qwen2.5-7B-Instruct (4-bit) on an M4 Pro.
- Profiled both engines and isolated why throughput stays **flat from 1 to 32 concurrent users**: a per-instance lock serializes requests inside each serving engine.
- Proved decode is memory-bandwidth-bound: K process-isolated instances behind one dispatcher return only **+9%** for **3× the memory** — a single instance already saturates the M4 Pro's ∼273 GB/s bus.

Open Source

Landed fixes in **ONNX Runtime** (ReshapeFusion crash, #33091) and **PyTorch** (ONNX int32-quantization export, #200180); open PR in **mlx-lm** (prompt-cache byte budget, #1944).

Education

BITS Pilani — M.Tech in Artificial Intelligence & Machine Learning, WILP 2025 – 2027 (expected)
Bhilai Institute of Technology, Durg — B.Tech in Computer Science Engineering 2019 – 2023